

GWU CAPSTONE · DATA SCIENCE · APRIL 2026

Prostate Cancer Epidemiology Dashboard

An Interactive Statistical Platform for Population-Level Analysis

Ava McClinton · SEER 22, NCI · 1999–2022 · 51 States + DC

Why This Project Matters

268K

new diagnoses
per year in the U.S.

#2

leading cancer
killer in U.S. men

1 in 8

men will be
diagnosed in their lifetime

Prostate cancer is the most common non-skin cancer in American men. Screening practices have shifted dramatically over the last decade, creating measurable changes in stage distribution at diagnosis — and yet racial disparities remain large and persistent.

This dashboard provides a comprehensive, interactive platform to explore 23 years of SEER data — built entirely as a self-contained HTML/JavaScript application with no server.

12 Interactive Charts

51 States · 23 Years

Race · Stage · Survival · Geography

Data Foundation

Primary Source: SEER 22

Surveillance, Epidemiology & End Results Program
National Cancer Institute (NCI)
Covers approximately 35% of the U.S. population
Gold standard for U.S. cancer epidemiology

Years covered: 1999–2022 (23 years of data)

Geography: 51 states + Washington D.C.

Rates: Age-adjusted to 2000 U.S. standard population

Metrics: Incidence, mortality, stage at diagnosis,
5-year survival, race/ethnicity breakdowns

Quality: High completeness, peer-reviewed methodology

What the Dashboard Contains

Trends

Incidence & mortality 2000–2022, stage-filtered, with linear regression projections to 2030

Disparities

Race/ethnicity analysis across 5 groups, lollipop charts, trend lines, and 2030 projections

Geography

50-state choropleth map, incidence vs. mortality toggle, state-to-state comparison tool

Survival

5-year relative survival rates, Sankey flow diagram, age distribution of cases

Your Risk

Personalized risk calculator using age and race with relative risk tier scoring

TECHNICAL ARCHITECTURE

How the dashboard was designed and built from scratch

System Architecture

HTML/CSS Shell

Structure & Styling
Playfair Display + Inter
Responsive tabs & KPI cards

Embedded Data

RAW JavaScript Object
SEER 22 · 23 years
All states, stages, races

Plotly.js (CDN)

3-Fallback Async Loader
plot.ly → jsDelivr → unpkg
Fault-tolerant rendering

Chart Functions

12 render() functions
Linear regression engine
Interactive state mgmt

✓ Self-contained: a single .html file that runs in any browser, online or offline, with no server, no database, no build step, and no configuration. Can be shared via email, USB drive, or GitHub Pages.

Portability

Single file — runs anywhere
without dependencies

Reliability

3-CDN fallback ensures charts
render even if one CDN fails

Performance

All data embedded as JS literals
— zero data network requests

Build Process — From Data to Dashboard

01 Raw Data

SEER 22 CSV
download from
NCI SEER*Stat

02 Structure Data

Format as JS
objects, clean
field names

03 HTML Shell

Tabs, KPI cards,
CSS styling,
responsive layout

04 Plotly Charts

12 render()
functions built
iteratively

05 Interactions

Filters, toggles,
map, risk calc,
state compare

06 Projections

Linear regression
+ 95% CI bands
for all groups

07 Bug Fixes

Stage filter,
KPI values,
annotation labels

08 Polish & Ship

Race labels,
axis titles,
GitHub commit

Key Code: Fault-Tolerant CDN Loader

```
var CDNS = [  
  'https://cdn.plot.ly/plotly-2.26.0.min.js',  
  'https://cdn.jsdelivr.net/npm/plotly...',  
  'https://unpkg.com/plotly.js-dist-min...'  
];  
  
window.onPlotlyReady = function(fn) {  
  if (window.Plotly) { fn(); }  
  else { _queue.push(fn); }  
};  
  
function tryLoad(i) {  
  if (i >= CDNS.length) return;  
  var s = document.createElement('script');  
  s.src    = CDNS[i];  
  s.onload = () => drainQueue();  
  s.onerror = () => tryLoad(i + 1); // retry  
  document.head.appendChild(s);  
}
```

Why 3 CDNs?

If `cdn.plot.ly` is down or blocked, the loader silently falls back to `jsDelivr`, then `unpkg`. The dashboard always loads charts.

Why a render queue?

Chart functions are called before Plotly finishes downloading. The queue holds them and drains automatically once Plotly is ready.

Why async? Why not `<script>`?

Synchronous script tags block the entire page from rendering. The async pattern lets the HTML and CSS paint immediately, then charts fill in.

Key Code: Projection Engine — OLS Regression

```
function linearRegression(xArr, yArr) {  
  const n      = xArr.length;  
  const sumX   = xArr.reduce((a,b) => a+b, 0);  
  const sumY   = yArr.reduce((a,b) => a+b, 0);  
  const sumXY  = xArr.reduce(  
    (s,x,i) => s + x*yArr[i], 0);  
  const sumX2  = xArr.reduce((s,x) => s+x*x, 0);  
  // Ordinary Least Squares slope & intercept  
  const slope  = (n*sumXY - sumX*sumY) /  
    (n*sumX2 - sumX*sumX);  
  const intercept = (sumY - slope*sumX) / n;  
  // Standard error → 95% CI bands  
  const residuals = yArr.map((y,i) =>  
    y - (slope*xArr[i] + intercept));  
  const se = Math.sqrt(residuals.reduce(  
    (s,r) => s+r*r, 0) / (n - 2));  
  return { slope, intercept, se };  
}
```

The Math Behind It

Ordinary Least Squares (OLS)

Finds the slope (β_1) and intercept (β_0) that minimize the sum of squared residuals across training years.

95% Confidence Interval

Upper band = $\hat{y} + 1.96 \times SE$

Lower band = $\hat{y} - 1.96 \times SE$

SE = std. error with $(n-2)$ degrees of freedom.

Training Windows

Overall trends: last 8 observed years.

Race projections: last 5 data points per group.

Projection horizon: 2023–2030.

 **Limitation:** Linear regression assumes past trends continue linearly. Projections do not account for new screening guidelines, treatment advances, or demographic shifts. Confidence bands widen as projection extends.

KEY FINDINGS

What 23 years of SEER data reveals about prostate cancer in the United States

Finding 1: Screening Policy Drives Incidence

2000–2012

Peak & Decline

Incidence fell from 175/100K (2000) to 90/100K (2014) — a 48% drop — as routine PSA screening fell out of favor after the 2012 USPSTF recommendation against it.

2018

Partial Restoration

Rates began rising after the 2018 update recommending shared decision-making for men aged 55–69. Localized-stage cases recovered as PSA testing resumed.

2008–2022

Distant Stage Up +50%

The most alarming trend: metastatic diagnoses at presentation rose steadily — from 8.4 to 12.6 per 100K — a likely consequence of reduced early screening.

Finding 2: Racial Disparities Are Not Closing

Race / Ethnicity	Incidence /100K	Mortality /100K	vs. White Men
Non-Hispanic Black	175.2	37.9	+60% incidence · +109% mortality
Non-Hispanic White	109.3	18.1	— baseline —
Hispanic	96.4	15.8	−12% incidence
Asian/Pacific Islander	60.3	9.2	−45% incidence

Projection Finding (New Chart)

Using OLS regression on 2014–2022 race-specific data, projections to 2030 show Non-Hispanic Black men remaining on a trajectory of approximately 185–195/100K — nearly 75% higher than the projected rate for White men (~115/100K). The racial gap does not narrow on the current trend. This is the central policy conclusion of this analysis.

Finding 3: Stage at Diagnosis Is Decisive

~100%

5-Year Survival
Localized Stage

76% of all cases
caught early

99.9%

5-Year Survival
Regional Stage

13% of all cases

38%

5-Year Survival
Distant / Metastatic

↑ 8.4 → 12.6 /100K
since 2008

The 62-point survival gap between localized and metastatic diagnosis is the single strongest argument for early screening. The Sankey diagram in the dashboard makes this concrete: out of 1,000 diagnosed men, 942 survive 5 years — but only because 76% are caught at the localized stage. If the distant-stage trend continues rising, this survival advantage will erode.

The rise in distant-stage cases (+50% since 2008) is the most urgent finding in this dataset.

Conclusion

What Was Built

Self-contained HTML dashboard (1,800+ lines)

12 interactive Plotly.js visualizations

6 thematic tabs with full interactivity

Fault-tolerant 3-CDN async Plotly loader

OLS projection engine with 95% CI bands

Race-specific incidence projections to 2030

50-state choropleth + state comparison tool

Personalized risk calculator (age $\times$ race)

Key Takeaways

Screening policy is the dominant driver

The 2012 USPSTF change caused a 48% incidence drop; the 2018 reversal is clearly visible in the trend data.

Racial disparities persist through 2030

Black men face 60% higher incidence and 109% higher mortality. Projections show the gap does not narrow.

Stage at diagnosis determines survival

~100% vs. 38% five-year survival. Distant-stage cases have risen 50% since 2008. Early detection is the key intervention.